GROUP 6

MHRS

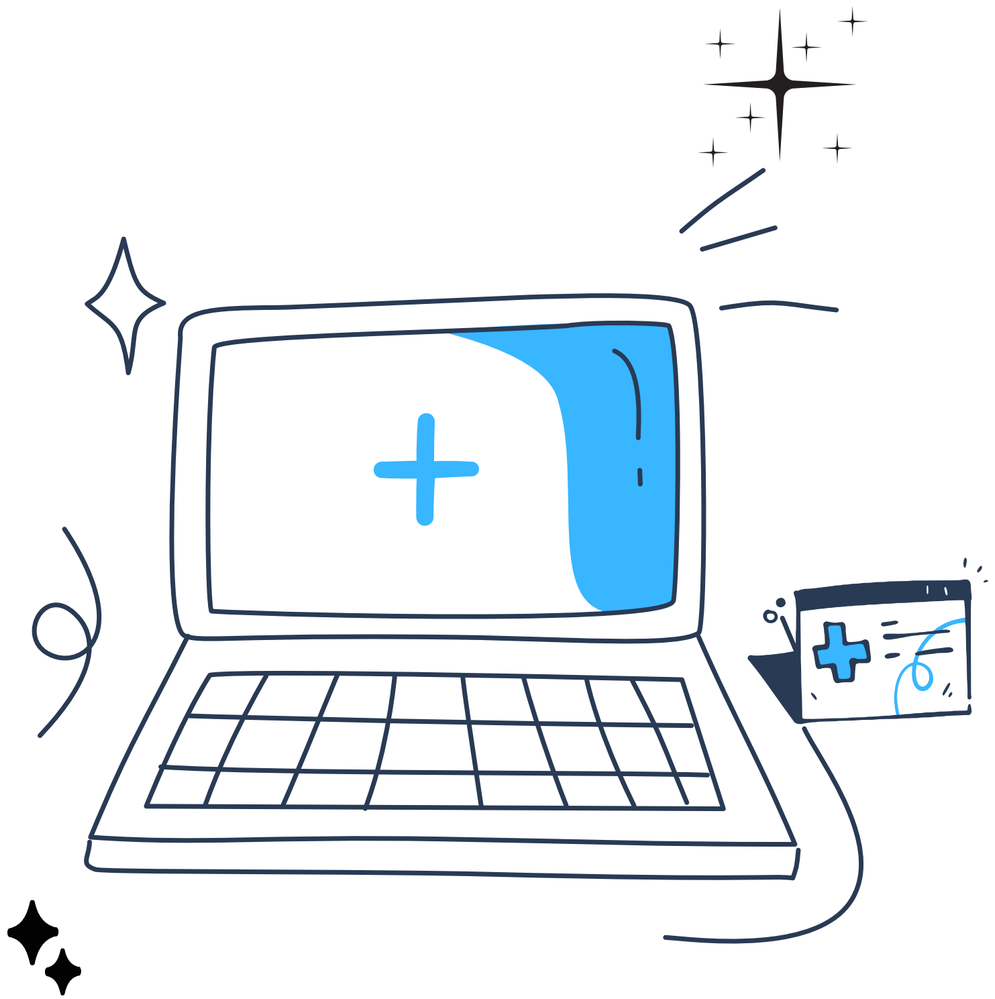
From Users to Measurements: Designing Usability Testing

WHERE WE ARE?

- Week 1 → System understanding
- Week 2 → Features & workflows
- Week 3 → User + Testing Design
- Week 4 → ...

WHY USABILITY MUST BE MEASURED?

- Usability is not a subjective opinion; it is a system metric.
- "Good design" must be backed by quantifiable human-computer interaction data.
- **Core Evaluation Pillars:**
  - **Effectiveness:** Accuracy and completeness of goals.
  - **Efficiency:** Resources expended in relation to accuracy.
  - **Satisfaction:** Comfort and acceptability of use.



DECONSTRUCTING USABILITY COMPONENTS

Effectiveness:
Can users actually complete the core tasks without critical failures?

Satisfaction:
What is the perceived friction and emotional response of the user?

Efficiency:
How much time and cognitive effort are?

Key Idea:
Every component will be tied to a specific, measurable variable.

QUANTIFYING THE USER EXPERIENCE:

1. **Task Success Rate (%):** Binary pass/fail metric for core workflows.

2. **Time on Task (Seconds):** Exact duration from task initiation to completion.

3. **Error Rate:** Count of misclicks, wrong navigations, and required recoveries.

4. **SUS Score (0-100):** System Usability Scale for standardized subjective evaluation.

5. **Qualitative Feedback:** Pain points captured through think-aloud observations.

OUR TESTING APPROACH

- **Task-Based Testing:** Users perform realistic scenarios in a controlled environment.
- **Think-Aloud Protocol:** Users verbalize their thought process while navigating.
- **Direct Observation:** Monitoring hesitations, mouse movements, and facial expressions.
- **Post-Test Survey (SUS):** Immediate structured evaluation after system interaction.
- **Key Principle:** Observe actual behavior, do not rely solely on user opinions.

TARGET AUDIENCE DIVERSITY

The Elderly User:
Low technology literacy, high risk of navigation errors, requires large UI elements and clear guidance.

The Working Adult:
Highly time-sensitive, expects maximum efficiency, low tolerance for multi-step friction.

The Caregiver/Parent:
Multi-tasking context, needs to manage appointments for dependents, uses complex system features.

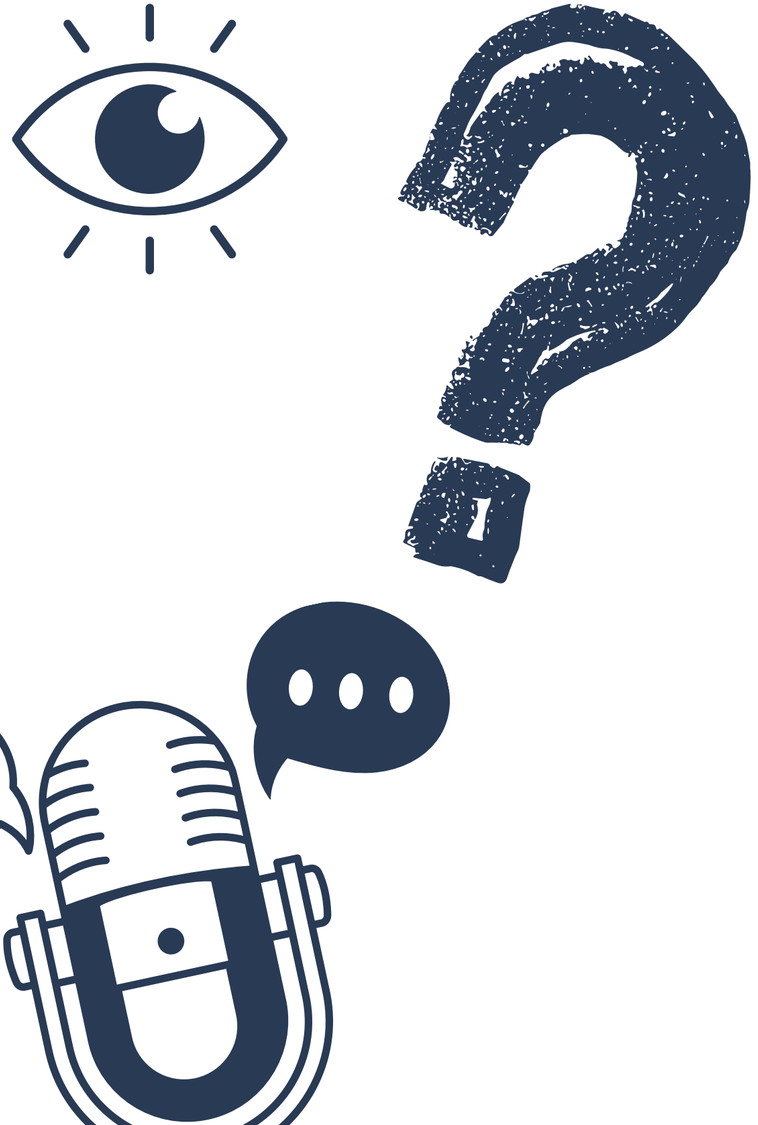
TEST PROTOCOL

Step-by-Step Execution Flow

- Phase 1: Introduction, briefing, and signing the consent form.
- Phase 2: Demographic and technical background questionnaire.
- Phase 3: Task execution utilizing the think-aloud protocol.

TEST PROTOCOL

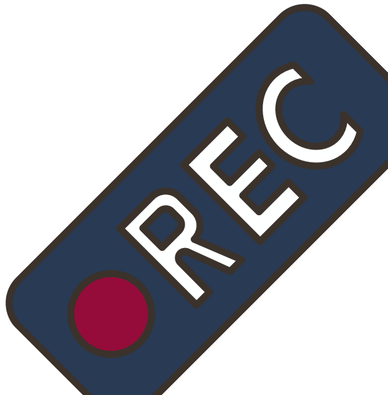
Step-by-Step Execution Flow



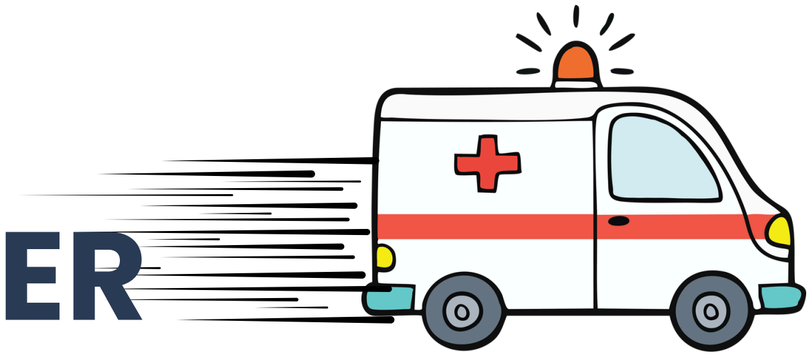
- Phase 4: Silent observation and active error logging by the moderator.
- Phase 5: Completion of the System Usability Scale (SUS) questionnaire.
- Phase 6: Short debriefing interview for final qualitative insights.

GATHERING MULTI-DIMENSIONAL DATA

- **Screen Recording:** Capturing cursor paths, clicks, and scrolling behavior.
- **Observation Sheets:** Standardized grids for logging hesitations and verbal cues .
- **System Error Logs:** Documenting dead ends or system unresponsiveness.
- **SUS Questionnaires**: Collecting normalized data for overall system rating.
- **Key Idea:** Merging quantitative metrics with qualitative behavioral insights.



WHAT WE WILL DELIVER



- Aggregated task success rates across different user personas.
- Average completion time baselines for core MHRS functions.
- A mapped frequency chart of the most critical UI/UX errors.
- An official SUS usability score representing the current system state.
- Next Step: Executing the design to collect real-world data in Week 4.

WHY THIS MATTERS?

The Impact of Structured Evaluation:

- Transforms subjective design critiques into objective, data-driven insights.
- Establishes clear baselines for future system improvements.
- Identifies exact friction points rather than general "bad design" complaints.
- Provides a scientifically sound foundation for a UI/UX redesign proposal.

SUMMARY

Week 3 Outcomes & Readiness:

- Usability: Shifted from a concept to a measurable target.
- Metrics: Strictly defined (Time, Errors, Success, SUS).
- Users: Modeled into distinct personas with varying tech literacy.
- Tasks: Structured into clear, real-world scenarios.
- Testing: Fully designed and protocolized.
- Final Statement: The framework is complete; we are ready for real user testing.

THANKS FOR LISTENING

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